

## NEW CARLISLE, QC, Canada

WMO: 716190

Lat: 48.0119N

Lon: 65.3317W

Elev: 46

StdP: 100.77

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |              |              |            |              | Coldest Month WS/MCDB |              |             |              | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|--------------|--------------|-------------------------------|------------|--------------|--------------|------------|--------------|-----------------------|--------------|-------------|--------------|-----------------------|------------|--------------|
|               |              |              | 99.6%                         |            |              | 99%          |            |              | 0.4%                  |              | 1%          |              |                       |            |              |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB         | DP           | HR         | MCDB         | WS                    | MCDB         | WS          | MCDB         | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-21.9 | (c)<br>-19.6 | (d)<br>-28.8                  | (e)<br>0.3 | (f)<br>-21.3 | (g)<br>-25.8 | (h)<br>0.4 | (i)<br>-19.0 | (j)<br>13.4           | (k)<br>-12.6 | (l)<br>12.2 | (m)<br>-12.2 | (n)<br>5.5            | (o)<br>290 | (p)<br>0.607 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.8                    | 26.2            | 19.9 | 24.5 | 18.7 | 22.9 | 17.9 | 21.1                | 24.6 | 20.0 | 23.1 | 19.0 | 21.7 | 3.3                  | 280  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 19.8                            | 14.6 | 23.0 | 18.8 | 13.7 | 21.6 | 17.9 | 12.9 | 20.7 | 61.5          | 24.7 | 57.6 | 23.1 | 54.4 | 21.7 | 25.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 10.5              | 9.2  | 8.0 | -24.8                      | 29.8 | 2.5                | 1.8 | -26.6                                              | 31.1 | -28.0      | 32.2 | -29.4      | 33.2 | -31.2      | 34.6 |
|                   |      |     | -25.2                      | 23.4 | 2.4                | 1.0 | -26.9                                              | 24.1 | -28.3      | 24.6 | -29.7      | 25.2 | -31.4      | 25.9 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 4.5         | -9.6  | -8.9  | -4.1  | 2.1   | 8.2   | 13.9  | 18.0  | 17.4  | 13.1  | 7.1   | 1.1   | -5.2  |  |  |
|                                                                        | DBStd       | 10.38       | 5.72  | 5.51  | 4.93  | 3.43  | 3.49  | 3.55  | 2.71  | 2.67  | 3.44  | 3.51  | 4.20  | 4.88  |  |  |
|                                                                        | HDD10.0     | 2746        | 607   | 529   | 436   | 238   | 78    | 6     | 0     | 0     | 9     | 103   | 267   | 471   |  |  |
|                                                                        | HDD18.3     | 5106        | 866   | 762   | 694   | 487   | 313   | 139   | 39    | 51    | 161   | 348   | 517   | 729   |  |  |
|                                                                        | CDD10.0     | 740         | 0     | 0     | 0     | 1     | 24    | 123   | 247   | 228   | 102   | 14    | 1     | 0     |  |  |
|                                                                        | CDD18.3     | 58          | 0     | 0     | 0     | 0     | 0     | 7     | 28    | 21    | 3     | 0     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 282         | 0     | 0     | 0     | 0     | 4     | 47    | 127   | 85    | 19    | 0     | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 39          | 0     | 0     | 0     | 0     | 0     | 6     | 18    | 13    | 2     | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 3.5         | 4.4   | 4.2   | 3.8   | 3.5   | 3.1   | 2.8   | 2.6   | 2.7   | 2.9   | 3.5   | 3.8   | 4.2   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 1040        | 79    | 64    | 71    | 72    | 88    | 90    | 102   | 97    | 90    | 104   | 92    | 91    |  |  |
|                                                                        | PrecMax     | 1317        | 120   | 128   | 152   | 163   | 188   | 162   | 152   | 168   | 177   | 202   | 156   | 188   |  |  |
|                                                                        | PrecMin     | 761         | 35    | 2     | 23    | 21    | 23    | 32    | 47    | 26    | 28    | 16    | 19    | 34    |  |  |
|                                                                        | PrecStd     | 139         | 28    | 27    | 31    | 34    | 40    | 34    | 34    | 36    | 40    | 52    | 32    | 38    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 5.4   | 5.2   | 8.7   | 16.3  | 23.6  | 27.3  | 29.0  | 28.5  | 25.9  | 19.3  | 13.4  | 6.7   |  |  |
|                                                                        |             | MCWB        | 5.1   | 4.0   | 4.6   | 9.2   | 14.8  | 19.6  | 21.7  | 21.3  | 19.8  | 14.7  | 12.4  | 6.2   |  |  |
|                                                                        | 2%          | DB          | 2.5   | 2.6   | 5.5   | 11.1  | 19.7  | 24.5  | 26.3  | 25.5  | 22.3  | 16.1  | 10.7  | 4.6   |  |  |
|                                                                        |             | MCWB        | 1.9   | 1.5   | 2.6   | 6.1   | 12.7  | 17.8  | 20.3  | 19.8  | 17.4  | 12.8  | 9.5   | 3.9   |  |  |
|                                                                        | 5%          | DB          | 0.6   | 0.8   | 3.7   | 8.8   | 16.9  | 22.1  | 24.5  | 23.7  | 20.2  | 14.0  | 8.7   | 2.9   |  |  |
|                                                                        |             | MCWB        | -0.1  | -0.3  | 1.7   | 5.2   | 11.6  | 16.5  | 19.1  | 18.6  | 16.7  | 11.7  | 7.2   | 1.9   |  |  |
|                                                                        | 10%         | DB          | -1.2  | -0.8  | 2.3   | 7.0   | 14.4  | 20.0  | 23.1  | 22.2  | 18.6  | 12.3  | 6.8   | 1.5   |  |  |
|                                                                        |             | MCWB        | -2.0  | -1.8  | 0.6   | 3.8   | 10.3  | 15.3  | 18.3  | 17.8  | 15.6  | 10.3  | 5.4   | 0.7   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 5.3   | 4.5   | 5.8   | 10.4  | 16.9  | 21.1  | 22.7  | 22.2  | 20.7  | 15.8  | 12.6  | 6.4   |  |  |
|                                                                        |             | MCDB        | 5.4   | 4.9   | 7.0   | 14.1  | 20.8  | 26.0  | 27.0  | 26.3  | 24.1  | 18.1  | 13.1  | 6.6   |  |  |
|                                                                        | 2%          | WB          | 2.0   | 1.7   | 3.5   | 7.5   | 14.1  | 19.1  | 21.4  | 20.7  | 18.8  | 13.7  | 10.0  | 4.2   |  |  |
|                                                                        |             | MCDB        | 2.4   | 2.3   | 4.7   | 9.8   | 17.6  | 22.7  | 24.9  | 24.1  | 20.9  | 15.4  | 10.6  | 4.5   |  |  |
|                                                                        | 5%          | WB          | 0.0   | -0.1  | 2.0   | 5.7   | 12.4  | 17.6  | 20.2  | 19.7  | 17.4  | 12.2  | 7.5   | 2.2   |  |  |
|                                                                        |             | MCDB        | 0.5   | 0.7   | 3.4   | 7.9   | 15.9  | 20.6  | 23.3  | 22.5  | 19.7  | 13.6  | 8.4   | 2.7   |  |  |
|                                                                        | 10%         | WB          | -1.9  | -1.8  | 0.8   | 4.3   | 10.7  | 16.1  | 19.2  | 18.7  | 16.0  | 10.6  | 5.6   | 0.8   |  |  |
|                                                                        |             | MCDB        | -1.2  | -0.8  | 2.2   | 6.5   | 13.9  | 19.1  | 22.1  | 21.1  | 18.1  | 12.0  | 6.5   | 1.4   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 7.5   | 7.9   | 7.7   | 7.3   | 8.8   | 9.3   | 8.8   | 8.8   | 8.8   | 7.0   | 5.9   | 6.2   |  |  |
|                                                                        |             | MCDBR       | 9.1   | 8.0   | 8.7   | 10.8  | 13.5  | 12.6  | 11.1  | 10.9  | 10.7  | 9.3   | 7.6   | 6.0   |  |  |
|                                                                        | 5% WB       | MCWBR       | 9.2   | 7.1   | 6.7   | 7.1   | 8.0   | 7.5   | 6.1   | 6.0   | 6.8   | 6.8   | 7.0   | 6.0   |  |  |
|                                                                        |             | MCWBR       | 9.2   | 7.4   | 7.8   | 9.2   | 11.9  | 11.0  | 9.6   | 9.5   | 8.4   | 7.2   | 6.2   | 6.2   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.275 | 0.279 | 0.313 | 0.382 | 0.368 | 0.386 | 0.422 | 0.394 | 0.361 | 0.334 | 0.299 | 0.279 |  |  |
|                                                                        |             | taud        | 2.362 | 2.301 | 2.301 | 2.240 | 2.380 | 2.354 | 2.303 | 2.379 | 2.453 | 2.508 | 2.513 | 2.427 |  |  |
|                                                                        | Ebn at Noon |             | 806   | 885   | 905   | 867   | 894   | 876   | 837   | 847   | 845   | 816   | 780   | 758   |  |  |
|                                                                        |             | Edh at Noon | 69    | 91    | 108   | 128   | 115   | 119   | 124   | 110   | 93    | 75    | 60    | 58    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1.26  | 2.08  | 3.32  | 4.34  | 5.13  | 5.54  | 5.44  | 4.77  | 3.68  | 2.28  | 1.23  | 0.91  |  |  |
|                                                                        | RadStd      |             | 0.13  | 0.22  | 0.26  | 0.36  | 0.44  | 0.43  | 0.43  | 0.30  | 0.30  | 0.18  | 0.15  | 0.09  |  |  |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |       | N/A     | N/A | +1.72  | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (15 neighbors) |       | N/A     | N/A | +1.38  | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page